AI Intelligence Platform V3 Installation and User Guide

System Requirements

- Python: Version 3.9 or higher is required.
- Operating Systems: Windows 10/11, macOS (Intel or Apple Silicon), or Linux (Ubuntu/Debian/CentOS).
- RAM: Minimum 8GB recommended.
- Internet: Active connection required for AI API calls and MongoDB synchronization.

Project Folder Preparation:

Ensure the following file structure is maintained in your project directory (ai_v3):

```
ai_v3/
├── app.py
├── .env
├── requirements.txt
├── historicdataCPU.csv
├── historicdataDISK.csv
├── historicdatamemory.csv
├── testdata.xlsx
└── README.md
```

Virtual Environment Setup

Virtual environments isolate the project dependencies to prevent conflicts with other Python projects.

Windows:

1. Open PowerShell or Command Prompt.
2. Navigate to your project folder: cd path\to\ai_v3
3. Create the environment: python -m venv venv
4. Activate the environment: venv\Scripts\activate

macOS and Linux:

1. Open the Terminal.
2. Navigate to your project folder: cd path/to/ai_v3
3. Create the environment: python3 -m venv venv
4. Activate the environment: source venv/bin/activate
5. Managing Dependencies (requirements.txt)

The requirements.txt file contains a list of every library the system needs to function.

Step 1: Create the requirements.txt file
If the file does not exist in your folder, create a new text file named requirements.txt and paste the following list into it:
streamlit
pandas
numpy
plotly
google-generativeai
openai
mistralai
pymongo
dnspython
openpyxl
python-dotenv

Step 2: Install the libraries
With your virtual environment activated, run the following command to install everything at once:
pip install -r requirements.txt

Note: If you add new features later and install more libraries, you should update this file using the command: pip freeze > requirements.txt

1. Configuration of Environment Variables (.env)

The system uses a .env file to store sensitive information like API keys. This prevents the keys from being hardcoded into the script.

1. Create a file named .env in the root folder.
2. Open it in a text editor (Notepad, VS Code, or TextEdit).
3. Add the following lines, replacing the placeholder text with your actual credentials:

MONGODB_URI=mongodb+srv://username:password@cluster.mongodb.net/
OPENAI_API_KEY=sk-xxxx
GOOGLE_API_KEY=AIza-xxxx
MISTRAL_API_KEY=xxxx

Data Source Verification

The system relies on local files for the initial dashboard and analysis:

- Server Health: historicdataCPU.csv and historicdataDISK.csv must contain a Date Time column and columns ending with (RAW).
- Inventory: testdata.xlsx must contain sheets named Inventory Work Sheet and Tickets for Analysis.

1. Running the Application
2. Ensure the virtual environment is active (you should see (venv) at the start of your command line).
3. Execute the application:
   streamlit run app.py
4. The terminal will output two URLs. Copy the Local URL (usually http://localhost:8501) into your web browser.
5. Detailed Usage Instructions

Module 1: Server Health Analysis

- The Analysis tab allows you to select a specific time window.
- The AI uses the CSV data filtered by your dates.
- By checking Include historical database context, the app queries your MongoDB cluster for the last 5 records of previous analyses. This allows the AI to recognize if a current CPU spike is a recurring event.

Module 2: Inventory Intelligence

- The Visual Dashboard tab provides real-time charts.
- The AI Data Assistant tab is a chat interface.
- You can ask questions like "Which device type has the most tickets?" or "Plot a bar chart of inventory items."
- If you check Use historical chat context, the AI will pull previous chat records from MongoDB to maintain continuity in the conversation.

1. Maintenance and Updates

To update the system when code changes are made:

1. Pull the new app.py file.
2. Update dependencies if any new ones were added: pip install -r requirements.txt
3. Restart the streamlit server by pressing Ctrl+C in the terminal and running streamlit run app.py again.

Common Troubleshooting

- Python not found: Ensure Python is added to your PATH during installation. On Mac/Linux, use python3 instead of python.
- ModuleNotFoundError: This means a library was not installed. Re-run pip install -r requirements.txt.
- MongoDB Timeout: Ensure your IP address is whitelisted in the MongoDB Atlas Network Security settings.
- API Quota: If the AI returns an error, check your billing status on the OpenAI, Google, or Mistral dashboards.